

Phase 4: Statistics & Business Applications (Month 3)

Goal: Apply statistics to business decisions using mathematical reasoning, business case exercises, and practical coding applications.

1. Descriptive vs Inferential Statistics

Descriptive Statistics

Used to summarize and organize data.

Examples: Average monthly revenue Most popular product category Customer satisfaction summary **Inferential Statistics**

Used to make predictions or conclusions about a population using sample data.

Examples: Predicting customer churn Estimating future sales demand Testing if a marketing campaign improved conversions **Mathematical Concepts** Population vs Sample Sampling distributions Probability theory Confidence intervals **Business Implications** Helps managers reduce uncertainty Improves forecasting accuracy Supports evidence-based strategic decisions

2. Mean, Median, Mode & Standard Deviation

Mean (Average)

Formula: $\text{Mean} = \frac{\text{Sum of values}}{\text{Number of values}}$

Median

Middle value when data is sorted.

Mode

Most frequent value in the dataset.

Standard Deviation

Measures how spread out data points are from the mean.

Mathematical Concepts Central tendency Variance and dispersion Outliers Data distribution **Business Implications** Identify unstable sales periods Measure operational consistency Detect unusual customer behavior

3. Probability & Hypothesis Testing

Probability

Probability measures the likelihood of an event occurring.

Hypothesis Testing

A statistical method used to determine whether there is enough evidence to support a business claim.

Key Terms Null Hypothesis (H0) Alternative Hypothesis (H1) p-value Significance level Type I and Type II errors

Business Implications Evaluate whether promotions increase revenue Test effectiveness of pricing strategies
Validate customer satisfaction improvements

4. A/B Testing

A/B testing compares two versions of a business strategy to determine which performs better.

Examples Testing two website landing pages Comparing email subject lines Evaluating different product prices
Mathematical Concepts Sampling Conversion rates Statistical significance Confidence intervals **Business Implications** Improves marketing ROI Optimizes user experience Reduces decision-making bias

5. Data-Driven Decision Making

Data-driven decision making uses data analysis rather than intuition to guide business strategies.

Process Collect data Clean and organize data Analyze patterns Interpret results Take action **Business Implications**
Improves strategic planning Enhances operational efficiency Supports long-term growth

Business Case Exercises

Case Study 1: Retail Sales Analysis

Scenario: A retail store recorded weekly sales for 10 weeks. Management wants to understand the average performance and variability in sales.

Concepts Used: Mean Median Standard Deviation Trend Analysis

Business Implication: Business Implication: High standard deviation indicates unstable revenue. The company may need better inventory forecasting or promotional planning.

Python Code Example:

```
import numpy as np

sales = [120, 135, 140, 150, 145, 160, 170, 180, 175, 190]

mean_sales = np.mean(sales)
median_sales = np.median(sales)
std_sales = np.std(sales)

print(mean_sales, median_sales, std_sales)
```

Case Study 2: Marketing Campaign A/B Test

Scenario: A company tested two email subject lines. Version A had a 12% conversion rate while Version B had a 17% conversion rate.

Concepts Used: Hypothesis Testing Conversion Rates Statistical Significance Probability

Business Implication: Business Implication: If Version B performs significantly better, the company should adopt it to improve customer acquisition and marketing efficiency.

Python Code Example:

```
from statsmodels.stats.proportion import proportions_ztest

success = [120, 170]
samples = [1000, 1000]

z_stat, p_value = proportions_ztest(success, samples)

print(z_stat, p_value)
```

Case Study 3: Customer Satisfaction Survey

Scenario: A business collected satisfaction scores from customers after introducing a new support system.

Concepts Used: Sampling Descriptive Statistics Confidence Intervals Inferential Statistics

Business Implication: Business Implication: Improved satisfaction scores indicate better customer retention and stronger brand loyalty.

Python Code Example:

```
import scipy.stats as stats
import numpy as np

scores = [4,5,5,4,3,5,4,4,5,5]

mean = np.mean(scores)
confidence = stats.t.interval(
    0.95,
    len(scores)-1,
    loc=mean,
    scale=stats.sem(scores)
)

print(confidence)
```

Key Statistical Formulas

Concept	Formula / Meaning
Mean	Mean = Sum of values / Number of values
Variance	Average squared distance from mean
Standard Deviation	Square root of variance
Probability	Favorable outcomes / Total outcomes
Confidence Interval	Range likely containing population parameter
Hypothesis Testing	Compare p-value with significance level

Final Takeaways

Statistics transforms raw data into actionable business insights. Descriptive statistics help explain what happened. Inferential statistics help predict what may happen. A/B testing enables evidence-based optimization. Data-driven organizations make more accurate and profitable decisions. Python and statistical tools automate business analysis efficiently.